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Al Isra Private University
Faculty of Engineering
Electrical Engineering Department

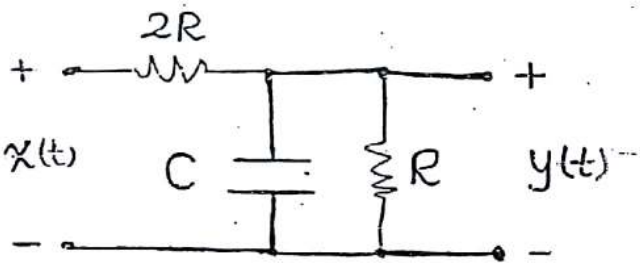
Student Name:_____
Student Number:_____

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Second Exam

Course Name:	Systems & Signals Processing	Date:	May 16, 2005
Course No:	402-322	ExamTime:	1
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Note: Answer only 4 problems of the 5 below.

1. Using Table 4-1 and Table 4-2, find the Fourier Transform of:
   a. $x(t) = 5.exp(-4t + 1).u(t)$.
   b. $y(t) = 4.sin(50\pi t).cos(500\pi t)$.
   c. $z(t) = 4/|2 + j\pi t|^2$.
2. The complex exponential Fourier Series over an interval $0 < t < T$ is given by $x(t) = \sum (4/(1 + jn))exp(j(5\pi nt - n\pi/8))$ for all n. Find:
   a. The average value. b. The fundamental frequency.
   c. The amplitude and the phase of the third harmonic.
   d. The average power contained in the first three harmonics.
3. Consider a fixed linear system with amplitude response $2.\Pi[w/600] + 2.\Pi[w/300]$ and phase response $-0.0025\pi\, w$ for $|w| < 200$, 0.5π for $w < -200$, and -0.5π for $w > 200$. Find the output and determine the type of distortion for the following inputs:
   a. $x_1(t) = cos(100t) + 2\, cos(200t)$. b. $x_2(t) = cos(180t) + cos(250t)$.
4. Using the integral form, find the inverse Fourier transform of the signal $X(f) = 4.exp(-2.f.sgn(f))$.
5. Using the Fourier transform theory, find the impulse response $h(t)$ of the circuit shown in Fig. 1.



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Q1 a. $x(t) = 5e^{-4t+1}u(t) = 5e^{-4(t+0.25)}u(t)$

(5) $X(f) = 5\dfrac{e^{-j2\pi f(0.25)}}{4+j2\pi f}$

b. $x(t) = 4\sin(50\pi t)\cos(500\pi t)$

$= -j[\delta(f-25) - \delta(f+25)] * [\delta(f-250) + \delta(f+250)]$

$= -j[\delta(f-275) - \delta(f+275) - \delta(f-225) + \delta(f+22$

c. $Z(t) = \dfrac{4}{|2+j\pi t|^2} = \dfrac{4}{4+(\pi t)^2} = 2\cdot\dfrac{2(4)}{4^2+(2\pi t)^2}$

$Z(f) = 2e^{-4|-f|} = 2e^{-4|f|}$

Q2 $x(t) = \sum \dfrac{4}{1+jn} e^{-j\frac{n\pi}{8}} e^{j5\pi nt}$

(6) $X_n = \dfrac{4}{1+jn} e^{-j\frac{n\pi}{8}}$

1a. The average value $= X_0 = 4$,

2b. $X_3 = \dfrac{4}{1+j3} e^{-j\frac{3\pi}{8}}$ $\quad |X_3| = \dfrac{4}{\sqrt{1+9}} = \dfrac{4}{\sqrt{10}}$

$\theta_3 = -\dfrac{3\pi}{8} - \tan^{-1}(3) =$

2d. $|X_1| = \dfrac{4}{\sqrt{1+1}} = \dfrac{4}{\sqrt{2}}$, $X_2 = \dfrac{4}{\sqrt{5}}$

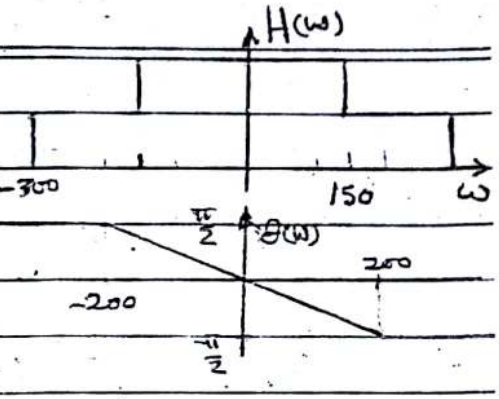
$P_{av} = |X_0|^2 + 2[|X_1|^2 + |X_2|^2 + |X_3|^2 = 16 + 2\left[\dfrac{16}{2} + \dfrac{16}{5} + \dfrac{16}{10}\right.$

$= 16 + 2\times 16\times\dfrac{8}{10} =$

1b. $\dfrac{D}{h} = \dfrac{5\pi}{2\pi} = 2.5$ rad.

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13 $|H(f)| = 2\Pi\left(\frac{\omega}{600}\right) + 2\Pi\left(\frac{\omega}{300}\right)$



5) $\theta(f) = \begin{cases} -0.0025\pi w & |w| \leq 200 \\ 0.5\pi & w < -200 \\ -0.5\pi & w > 200 \end{cases}$

a $x_1(t) = \cos(100t) + 2\cos(200t)$

$y_1(t) = 4\cos(100t - 0.25\pi) + 4\cos(200t - 0.5\pi)$

amplitude distortion

b. $x_2(t) = \cos(180t) + \cos(250t)$

$y_2(t) = 2\cos(180t - 0.45\pi) + 2\cos(250\pi t - \frac{\pi}{2})$

Phase distortion

24 $X(f) = 4e^{-2f\,sgn(f)}$

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$$x(t) = \int_{-\infty}^{0} 4e^{2f} e^{j2\pi ft}\,df + \int_{0}^{\infty} 4e^{-2f} e^{j2\pi ft}\,df$$

$$= \int_{-\infty}^{0} 4e^{(2+j2\pi t)f}\,df + \int_{0}^{\infty} 4e^{-(2-j2\pi t)f}\,df$$

$$= \frac{4}{(2+j2\pi t)} + \frac{4}{(2-j2\pi t)} = \frac{16}{4+(2\pi t)^2}$$

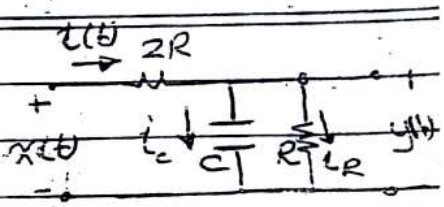
$$= 4\,\frac{2(2)}{(2)^2 + (2\pi t)^2}$$

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$i_R(t) = y(t)/R$

$i_c(t) = C\frac{dy}{dt}$



$i(t) = i_R(t) + i_c(t)$

$-x(t) + 2R\,i(t) + y(t) = 0$

$2RC\frac{dy}{dt} + 3y(t) = x(t)$

$Y(f)[3 + 2RC\,j2\pi f] = X(f)$

$H(f) = \frac{Y(f)}{X(f)} = \frac{1}{[3 + 2RC\,j2\pi f]}$

$= \frac{1/2RC}{\frac{3}{2RC} + j2\pi f}$

$h(t) = \frac{1}{2RC} \cdot e^{-\frac{3t}{2RC}} u(t)$.

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